

Multistep Equations with Rational Coefficients

Answer Key

Algebra 1

Quick Answers

- | | |
|-------|------------------------------------|
| 1. 28 | 7. 24 |
| 2. 15 | 8. 6 |
| 3. 20 | 9. 4 |
| 4. 12 | 10. 20 |
| 5. 10 | 11. 30 |
| 6. 8 | 12. (a) solution = $\emptyset$, — |
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What is verified

11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-REI.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 13 tagged checks.

1. Solve $\frac{x}{4} + 3 = 10$.

Solution: Subtract 3 from both sides: $\frac{x}{4} = 7$. Multiply both sides by 4: $x = 28$.

Check: $28 \div 4 + 3 = 10$.

$x = 28$

2. Solve $\frac{2}{3}(x - 9) = 4$.

Solution: Multiply both sides by $\frac{3}{2}$ (or by 3 then divide by 2), which undoes the fraction in front of the bracket: $x - 9 = 6$. Add 9 to both sides. Check: $\frac{2}{3}(15 - 9) = \frac{2}{3} \cdot 6 = 4$.

$x = 15$

3. Solve $\frac{3x}{4} = \frac{x}{2} + 5$.

Solution: The denominators are 4 and 2, so multiply every term on both sides by 4: $3x = 2x + 20$. Subtract $2x$ from both sides: $x = 20$. Check: $\frac{3(20)}{4} = 15$ and $\frac{20}{2} + 5 = 15$.

$$\boxed{x = 20}$$

4. Solve $\frac{x}{2} - \frac{x}{6} = 4$.

Solution: Multiply both sides by the common denominator 6: $3x - x = 24$, so $2x = 24$ and $x = 12$. (Combining first also works: $\frac{x}{2} - \frac{x}{6} = \frac{3x-x}{6} = \frac{x}{3}$.) Check: $6 - 2 = 4$.

$$\boxed{x = 12}$$

5. Solve $0.4(x + 5) = 6$.

Solution: Divide both sides by 0.4: $x + 5 = 15$. Subtract 5 from both sides. Distributing first gives the same answer: $0.4x + 2 = 6$, so $0.4x = 4$ and $x = 10$. Check: $0.4 \times 15 = 6$.

$$\boxed{x = 10}$$

6. Solve $\frac{2x - 1}{5} = 3$.

Solution: The whole numerator is divided by 5, so multiply both sides by 5 before touching anything else: $2x - 1 = 15$. Add 1 to both sides for $2x = 16$, then divide by 2. Check: $(16 - 1) \div 5 = 3$.

$$\boxed{x = 8}$$

7. Solve $\frac{3x}{4} - 2 = \frac{x}{6} + 12$.

Solution: The denominators are 4 and 6, so the common denominator is 12. Multiply every term on both sides by 12:

$$9x - 24 = 2x + 144$$

$$7x = 168$$

$$x = 24.$$

The -2 and the $+12$ must be multiplied by 12 as well — that is the step most often missed. Check: $\frac{3(24)}{4} - 2 = 18 - 2 = 16$ and $\frac{24}{6} + 12 = 4 + 12 = 16$.

$$\boxed{x = 24}$$

8. Solve $\frac{8x + 12}{4} = x + 9$.

Solution: Divide the whole numerator by 4 first: $2x + 3 = x + 9$. Then subtract x from both sides for $x + 3 = 9$, and subtract 3. Check: $(48 + 12) \div 4 = 15$ and $6 + 9 = 15$.

$$\boxed{x = 6}$$

9. Solve $\frac{x + 2}{3} - \frac{x - 4}{6} = 2$.

Solution: Multiply every term by the common denominator 6. The second fraction is subtracted, so its whole numerator changes sign:

$$2(x + 2) - (x - 4) = 12$$

$$2x + 4 - x + 4 = 12$$

$$x + 8 = 12 \Rightarrow x = 4.$$

Check: $\frac{6}{3} - \frac{0}{6} = 2 - 0 = 2$.

$x = 4$

10. Solve $0.25(x - 8) = 0.5x - 7$.

Solution: Distribute on the left: $0.25x - 2 = 0.5x - 7$. Subtract $0.25x$ from both sides: $-2 = 0.25x - 7$. Add 7 to both sides: $5 = 0.25x$. Divide both sides by 0.25, which is the same as multiplying by 4. Check: $0.25(20 - 8) = 3$ and $0.5(20) - 7 = 3$.

$x = 20$

11. Solve $\frac{2}{5}(x + 10) - \frac{x - 6}{2} = 4$.

Solution: Common denominator 10; multiply every term on both sides by it:

$$4(x + 10) - 5(x - 6) = 40$$

$$4x + 40 - 5x + 30 = 40$$

$$-x + 70 = 40 \Rightarrow -x = -30 \Rightarrow x = 30.$$

Check: $\frac{2}{5}(40) - \frac{24}{2} = 16 - 12 = 4$.

$x = 30$

12. $\frac{3x - 1}{2} + \frac{x}{4} = \frac{7x}{4} + 1$: (a) solve; (b) what does the last line mean?

Solution (a): Multiply every term by the common denominator 4:

$$2(3x - 1) + x = 7x + 4$$

$$6x - 2 + x = 7x + 4$$

$$7x - 2 = 7x + 4$$

$$-2 = 4.$$

The variable terms are identical on both sides and cancel completely, leaving a statement with no x in it at all — and a false one.

no solution

Solution (b) — instructor-judged. Because $-2 = 4$ is false, there is no number that can be substituted for x to make the original equation true: the two sides differ by a fixed amount of 6 whatever x is. The solution set is empty. Full credit says the false numerical statement means no value of x works; writing $x = 0$, or treating $-2 = 4$ as something to solve, does not.